An Electrical Power System of Mars Rover

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***Abstract*-This paper focuses on the design and construction of a Mars rover electrical power system, which support the exploring equipments working over the day and night. The electrical power system is a direct energy conversion system consisting of solar panel, Li-ion batteries and radioisotope thermoelectric generator. PV cells are used as primary power source; batteries as emergency power source, and RTG as auxiliary power source. Thus, the implementation of an energy management system is required, which is capable of predicting future power generation, planning activities within expected power resources range, analyzing the healthy state of power system, and diagnosing power system failures and emergencies.**

***Keywords*-electrical power system, Mars rover, energy management**

I. INTRODUCTION

The future chinese Mars rover consists of nine systems, which are mobility system, structure and mechanism system, navigation and control system, integrative electronics system, electrical power system, thermal control system, TT&C and data transmission system, manipulator, payload[1].

The electrical power system is one of the key systems of the Mars rover. In order to generate, store, transfer, and distribute energy, the electrical power system needs to be able to supply reliable and stabile power over both days and nights, drive the wheel motors, support the survey activity on the red planet, help the electronic equipments survive from the extreme cold night, and awake Mars rover from sleeping mode. The planet of Mars is approximately 97,131,000 km further away from Sun than Earth is. Because of its larger orbit radius, the length of a Martian year is equal to 1.88 Earth year. Temperatures on Mars range from -140℃ in winters to +20℃ in summers. There is a variety of ways of powering a Mars rover, such as batteries, RTG, and solar cells. Up to now, the successful Mars rovers use solar panels, li-ion batteries, and RTG altogether as power source. This paper presents an electrical power system for future Mars rover which will land and survey on Mars. This system uses direct energy conversion topology, and consists of two solar panels on the top, two Li-ion secondary batteries and one radioisotope thermoelectric generator. An energy management system is also designed to predict future power generation, plan activities within expected power resources range, analyze power system state and health, and diagnose power system failures and emergencies.

II. ELECTRICAL POWER SYSTEM PROJECT

The power system on the Mars rover is a direct energy transfer (DET) system. The primary source of power is the solar array which consists of two deployable panels mounted on the top of rover structure. Power from the solar panels is directly transferred to the spacecraft bus, which is regulated in +29 ± 1volts. Two sets of rechargeable Li-ion batteries are selected as energy storage device, which possesses the ability to provide power at least 90 Mars days on the surface of Mars and operate over a wide temperature range (–20℃ to +40℃). The RTG works as auxiliary power source to meet power consumption of heating during the sleep mode. The Mars rover electrical power architecture has been designed as shown in the following figure 1.

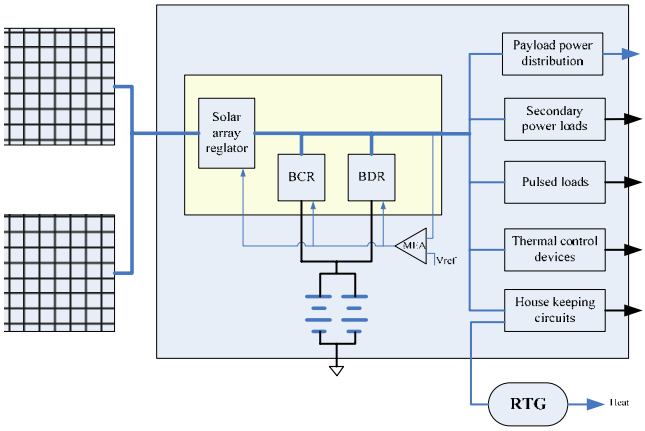


Fig.1 electrical power architecture of Mars rover

A. Solar panel

Solar cells for the Mars surface shall withstand a severe environment as the seasons changed, characterised by dust storm, low temperature, low solar intensity and by a solar spectrum that changes daily and seasonally. The spectrum on Mars surface has a low blue component affected by the atmosphere opacity[2].

The analysis of article[3] shows that for the existing available solar cells, Si, as dual junction GaAs/Ge solar cells, and triple junction GaAs/Ge solar cells, the triple-junction GaAs/Ge solar cell are still the best candidate among the existing available solar cells. The efficiency of standard triple-junction GaAs/Ge solar cells can reach 22% under all conditions of Mars where the spectrum is blue-poor. The solar cells are tested under a

solar intensity range from 30% to 60% of AM0, and a temperature range from -120°C to +20 °C. Triple junction solar cells show efficiency in the range from 21% under very low intensity illumination to 31% under cold, low optical depth conditions[4].



Fig.2 profile of triple junction solar cell

The rover employs two deployable panels mounted on the top of rover structure, Each array panel has a honeycomb-type construction.

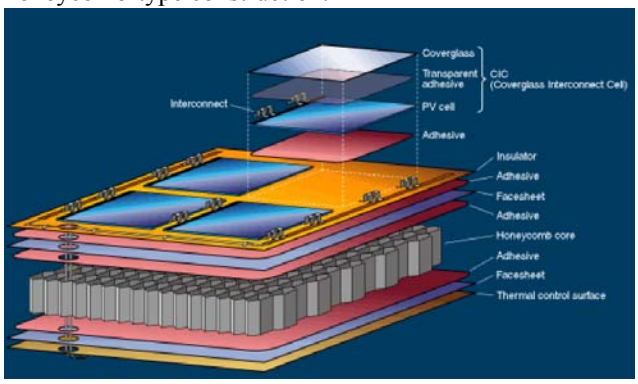


Fig.3 structure of solar panel

Solar panel Construction：

• Construct panel with 21 cells in series to provide the required voltage.

• 128 Parallel strings provide required current.

• meet power demand for minimum performance requirements.

B. Lithium-ion batteries

Lithium Ion cells and batteries are widely used in aerospace applications nowadays because of their high power density, specific energy and energy density. The two successful Rovers landed on Mars, Spirit and Opportunity of JPL, NASA are both equipped with rechargeable Lithium-ion batteries [5].

Compared with other kinds batteries, new type Lithium-ion batteries posses the following advantages: a) higher specific energy and energy density, b) long cycle life and calendar life, c) improved low temperature performance, d) low self-discharge, e) high charge/discharge efficiency.

TABLE Ⅰ

CHARACTERISTICS OF RECHARGEABLE BATTERIES

Technology	Specific Energy Wh/kg	Energy Density Wh/l	Operating Temp. Range, ℃	Cycle life
Ag-Zn	90	191	-20 to 25	>100
Ni-Cd	34	150	-10 to 25	>25k(25%DOD)
Ni-H2	60	90	-10 to 25	>30k(25%DOD)
Li-Ion	90	250	-20 to 30	>500 (50% DOD)

Lithium-ion rechargeable batteries are selected in order to meet the tight mass and volume constraints of the development. The required battery capacity is 40Ah. This is achieved through 4 parallel-connected cell strings with 7 cells in series (7 series cells per string and 4 strings in parallel) module, which providing 40Ah at a maximum terminal voltage of 29.4V.

C. Radioisotope thermoelectric generator

The electrical power system of the Mars Rover should work well under the entire temperature range of the Martian environment, which can vary from a low of -140℃ to a high of +20℃ during the Martian night and day. The electrical power also uses a radioisotope thermoelectric generator (RTG) in order to generate the power as auxiliary power for rover's needs. The RTG has a long lifetime and enough waste heat to maintain proper temperature for the electronics inside the body of the rover. The thermal waste heat is pumped into the rover via a fluid loop. The electronic system working under constant temperature can use a combination of waste heat and electrical heaters to survive the extreme cold temperatures of Mars. RTG enables the rover to operate during the day and maintain the temperature for electronic equipments working properly in the night.

The technology is a finite energy source that has scalable energy production. The process is based on the decay of polonium in order to produce energy in a nuclear mode. After the heat energy is produced by the RTG, it is then converted to electrical energy through a thermocouple. The simplest nuclear power systems used in space are Radioisotope Heater Units (RHUs). These devices contain an amount of radioactive material to generate heat directly via radioactive decay.

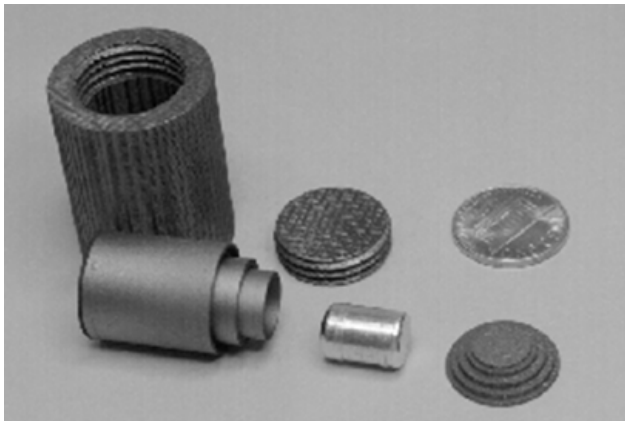


Fig.4 USA Light-Weight RHU - 1Wt - $^{238}PuO_2$



Fig.5 Image of a mostly thermally-isolated plutonium RTG pellet glowing red hot[6]

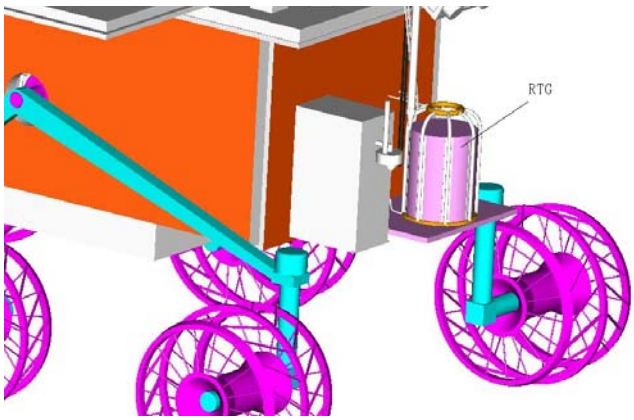


Fig.6 diagram of RTG fixed on the Mars rover

The RTG use radioactive fuel polonium-238 with outputting power of 1.8W, and thermal power of 80W. It cannot support the rover survey mission at night on the Mars, for the power and heat can only maintain the temperature of electronics under sleep mode. The RTG vessel is arranged at the back of the rover.

D. Power control unit

The Power bus voltage is controlled by a conventional three-domain control system, and the regulated bus voltage is +29±1V. During sunlight period, the power is delivered by the solar panel through a Sequential Switching Shunt Regulator (S3R), which is controlled by a fully reliable Main Error Amplifier (MEA). Batteries are controlled by the Battery Discharge Regulators (BDR) and Battery Charge Regulators (BCR). The BDR use voltage boost regulators with over current/voltage protection. During daytime, batteries are charged with constant current by BCR, including two nominal charge rates and constant voltage charging. The batteries are charged at constant regulated current, until a setted voltage limit is reached. The BCR will then keep the battery at this voltage level. When output power of the solar panel exceed the total load demand, including battery charging, the shunt regulator will short unnecessary sections of the solar array. This will regulate the bus voltage according to the power demand and keep the bus voltage within a small range.

E. Energy management system

The energy management is a decision making process, which is designed to meet the load power demand with available power while optimizing the system energy usage. As the exploring mission is complicated, the Mars rover needs an energy management system to manage the power supply according to the survey plan scheduled for the rover. The energy management system must predict future power generation of rover, plan activities within expected power source, monitor and analyze the healthy state of power system, diagnose and respond to power system failures and emergencies [7]. the most important thing is that the energy stored should be equal to the energy consumed so that it can maintain the energy balance.

1. Autonomous management of power

Power management is supported by an adequate measurement of the power parameters. This includes array current and voltage, BDR output current, battery charge and discharge currents, total main bus current and voltage. We can identify the status of power supply and predict power generation by analyzing power parameters,. The power for each equipment can be regulated according to the task requirements and its priority so as to meet the limitation of power generation. The main task of autonomous management system includes monitoring the status of equipment on board, fault detection and isolation, generating fault isolation list for the reference to ground stuff.

2. Load Priority Management:

The scheduling of power is responsible for planning the schedules when loads will be turned on and off. The schedule is prepared according to the requests of the task command sequences. It has to maximize the execution of task command sequences request while adhering to the constraints of available power on the rover.

When there is a power overload, some loads will be removed from the power bus in order to make the balance

between energy generated and energy consumption. Loads with a lower priority will be removed before loads of higher priority. The load priority list ranks loads depending on whether they are considered essential or nonessential to the mission. When an overload occurs, the nonessential loads are continually disconnected until the overload is overcome.

3. Dynamic power scheduling and distribution

The fundamental principle of dynamic power scheduling and distribution is that the power consumption of loads must match the ability of power supply. The aim of Dynamic power scheduling is based on one mission period, such as take one day or one year as a management period is to provide a set of power scheduling rules so as to manage the power supply of loads, to cut off the lower priority payload, as lowering down the power consumption, preventing battery from over discharging and achieving the energy balance. The strategy not only make the charged/discharged energy reaching balance in each day and night period, but also sufficiently utilize the power provided by the solar panels.

4. Power System Fault Diagnose and Recovery:

Faults of power system of spacecraft can be divided into hard faults and soft faults. The former one refers to the faults which are relatively easy to be identified are caused by un-recoverable damages in components of power system. By contrary, the latter one is related to the changing of performance of power system's components, as solar array's degrading, parameter shifting of certain circuit. Generally speaking, soft faults are hard to be identified.

Fault management of power system generally means the detection, isolation, dealing and recovering of faults. In order to detect and isolate fault in time, the detection of faults must be carried out immediately and accurately, only in this way, the safety of loads and power bus can be ensured. As to the fault analysis and system recovery, they can be done in a few seconds without impacting explore mission. Yet, the forecasting of power system status would take tens of minutes for a great amount of data need to be computerized.

III. CONCLUSIONS

The electrical power system scheme of Mars rover uses solar panel, RTG and secondary li-ion batteries to meet the mars explore task power requirements. This paper demonstrates that two triple-junction deployable solar panel can support the rover surface operations, the batteries are designed to supply power in the launch and have the ability to provide power during night in at least 90 Mars days on the surface of Mars, the RTG provides enough power and heat to maintain the sleep mode of rover.

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